

```
Processando config 1, seed 0, teste: Rabbit
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Rabbit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 2, n = 8388608, r = 0, s = 32, L = 38,
Sparse = TRUE
```

```
Number of bits = n = 8388608
Number of cells =  $2^L$  = 274877906944
Expected number per cell =  $1 / 32768$ 
EColl =  $n^2 / (2k)$  = 128
Hashing = TRUE
```

```
Collision test
CollisionOver: density =  $n / k = 1 / 32768.00$ 
Expected number of collisions =  $\mu = 128.00$ 
```

```
-----
Results of CollisionOver test:
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```
POISSON approximation :
Expected number of collisions =  $N \cdot \mu$  : 256.00
Observed number of collisions : 254
p-value of test : 0.53
```

```
-----
Total number of cells containing j balls
```

```
j = 0 : 549739036926
j = 1 : 16776708
j = 2 : 254
j = 3 : 0
j = 4 : 0
j = 5 : 0
```

```
-----
CPU time used : 00:00:04.62
```

```
Generator state:
16777216 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
```

snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 16.00
p-value of test : 0.86

CPU time used : 00:00:00.05

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 11.00
p-value of test : 3.1e-5 *****

CPU time used : 00:00:00.07

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : 1.00
p-value of test : 0.16

CPU time used : 00:00:00.01

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 1.68
p-value of test : 0.98

Normal statistic for number of jumps : -51.64
p-value of test : 1 - eps1 *****

CPU time used : 00:00:00.05

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : 0.63
p-value of test : 0.26

CPU time used : 00:00:00.41

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : -0.099
p-value of test : 0.54

CPU time used : 00:00:00.21

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.031
p-value of test : 0.37

Kolmogorov-Smirnov- statistic = D- : 0.029
p-value of test : 0.42

Anderson-Darling statistic = A2 : 0.60
p-value of test : 0.64

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 5.59
p-value of test : 0.13

Global longest run of 1 : 22.00
p-value of test : 0.63

CPU time used : 00:00:00.07

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 115.01
p-value of test : 0.07

CPU time used : 00:00:00.01

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 28.45
p-value of test : 0.16

CPU time used : 00:00:00.01

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.30
p-value of test : 0.38

CPU time used : 00:00:00.01

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : -1.60
p-value of test : 0.95

CPU time used : 00:00:00.01

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : 1.39
p-value of test : 0.08

CPU time used : 00:00:00.01

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 196.46
p-value of test : 0.15

CPU time used : 00:00:00.01

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 284.35
p-value of test : 0.80

CPU time used : 00:00:00.01

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 2395.55
p-value of test : eps *****

CPU time used : 00:00:00.01

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1

```
-----
Normal statistic          :    0.72
p-value of test          :    0.24
-----
```

```
CPU time used            : 00:00:00.01
```

```
Generator state:
269499904 bits have been read.
```

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*****
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```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
```

```
sstring_AutoCor test:
```

```
-----
N = 1, n = 16777154, r = 0, s = 32, d = 2
-----
```

```
-----
Normal statistic          :    1.05
p-value of test          :    0.15
-----
```

```
CPU time used            : 00:00:00.01
```

```
Generator state:
286277088 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin
```

```
sstring_Run test:
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```
-----
N = 1, n = 3355443, r = 0, s = 32
-----
```

```
-----
Total number of 1 runs: 3355443
```

```
Number of degrees of freedom : 36
Chi2 statistic for number of runs : 39.78
p-value of test : 0.31
-----
```

```
-----
Total number of bits: 13418048
```

```
Normal statistic for number of bits : -0.72
p-value of test : 0.76
```

CPU time used : 00:00:00.05

Generator state:
299695136 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 51.57
p-value of test : 3.7e-11 *****

CPU time used : 00:00:00.05

Generator state:
316472352 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.03

Generator state:
333163552 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42	
ChiSquare statistic	:	979.29	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	42	
ChiSquare statistic	:	474.97	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	64	
ChiSquare statistic	:	212.31	
p-value of test	:	eps	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	36	
ChiSquare statistic	:	50.80	
p-value of test	:	0.05	

Test on the values of the Statistic C

Number of degrees of freedom	:	22	
ChiSquare statistic	:	22.81	
p-value of test	:	0.41	

CPU time used : 00:00:00.05

Generator state:
349940768 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94	
ChiSquare statistic	:	517.74	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	94	
ChiSquare statistic	:	424.59	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	512	
ChiSquare statistic	:	623.55	
p-value of test	:	5.2e-4	*****

Test on the values of the Statistic R

Number of degrees of freedom	:	90	
ChiSquare statistic	:	116.67	
p-value of test	:	0.03	

Test on the values of the Statistic C

Number of degrees of freedom	:	50	
ChiSquare statistic	:	51.69	
p-value of test	:	0.41	

CPU time used : 00:00:00.03

Generator state:

366717984 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed0.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	147.77
p-value of test	:	0.14

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	152.15
p-value of test	:	0.10

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	197.05
p-value of test	:	0.03

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	167.53
p-value of test	:	0.01

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	80.57
p-value of test	:	0.73

CPU time used : 00:00:00.03

Generator state:

383494784 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:06.10
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

Test	p-value
-----	-----
3 ClosePairsBitMatch, t = 4	3.1e-5
5 LinearComp	1 - eps1
17 HammingIndep, L = 64	eps
21 MatrixRank, 32 x 32	3.7e-11
22 MatrixRank, 320 x 320	eps
24 RandomWalk1 H	eps
24 RandomWalk1 M	eps
24 RandomWalk1 J	eps
25 RandomWalk1 H (L = 1024)	eps
25 RandomWalk1 M (L = 1024)	eps
25 RandomWalk1 J (L = 1024)	5.2e-4
-----	-----

All other tests were passed